

Emotion-Informed MBTI Prediction

Group 66 · AC 209b / CS 109b

Spring 2026 · Harvard University

Harry Hu · Tom Shan · Wendy Wang · Kemeng Zhang



depressed
people on
Instagram



depressed
people on
Reddit

Can your Reddit posts reveal your personality type?



u/Pavlos

I enjoy talking to my students. Based on the feedback from my students I know I'm the best professor. It makes me feel so happy. I'm more motivated to complete my grading TODOs.

E ↔ I

E?

N ↔ S

S?

F ↔ T

F?

J ↔ P

J?

Why This Matters



HR Screening

Automated assessment for cultural fit.



Recommendations

Personalized content via profiles.



Adaptive Learning

Tailoring pathways to styles.

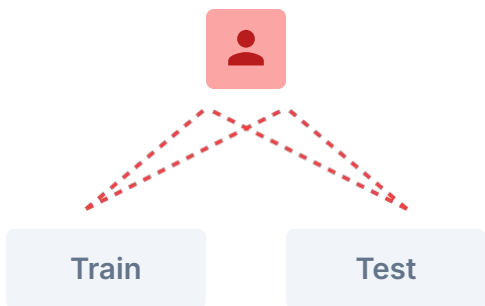
Research Questions:



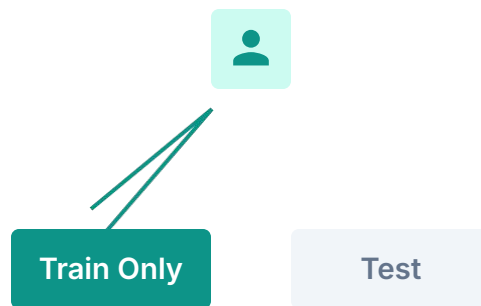
- How much **can text reveal** vs. pure noise?
- Do **emotion features** add signal beyond words?

The Problem with Prior Work

Post-Level Splits



Author-Level Splits



Post-level splits let models memorize writing style — we predict per author instead.

Data at a Glance

13M

Raw Posts



1.65M

Filtered Posts



10,414

Authors

Reddit MBTI Corpus

"I love exploring new ideas..."

Label: ENFP

"Logic is the key to solving..."

Label: INTP

Emotion-Balanced Corpus

"I am so thrilled about this!"

Label: Joy

"This makes me incredibly upset."

Label: Anger

Filtered by quality (5+ words/post, 20+ posts/author) with a 200-post cap per author.

Leakage Masking

Before Masking

"As an INTJ, I find that logic is essential for..."

After Masking

"As an [TYPE], I find that logic is essential for..."

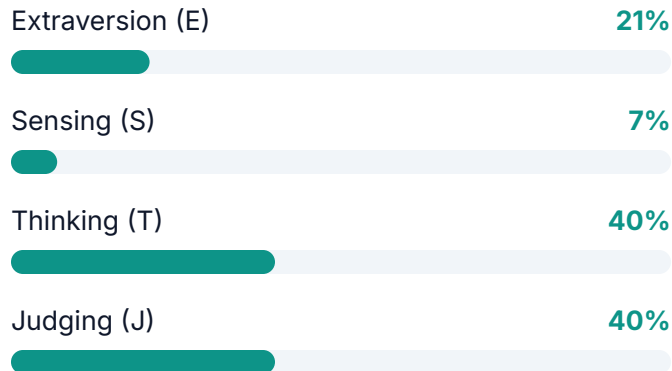
Authors mentioning type in posts **78%**

Leakage after masking **0%**

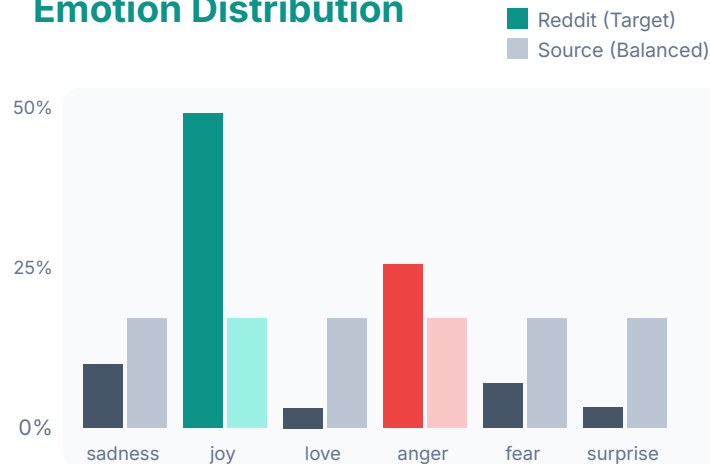
We mask all MBTI-related tokens with a placeholder to prevent trivial model shortcuts.

Class Imbalance & EDA

MBTI Dimension Distribution

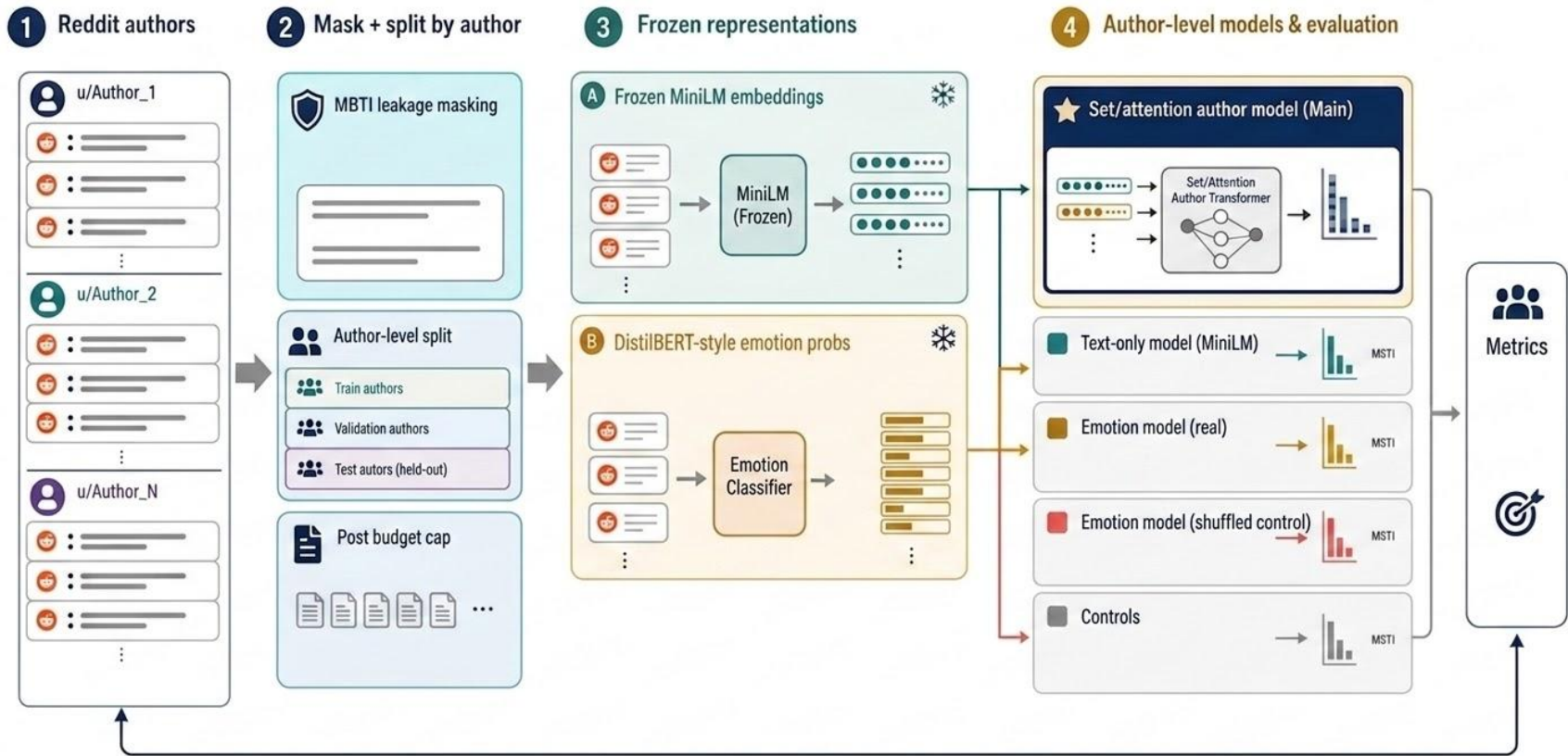


Emotion Distribution

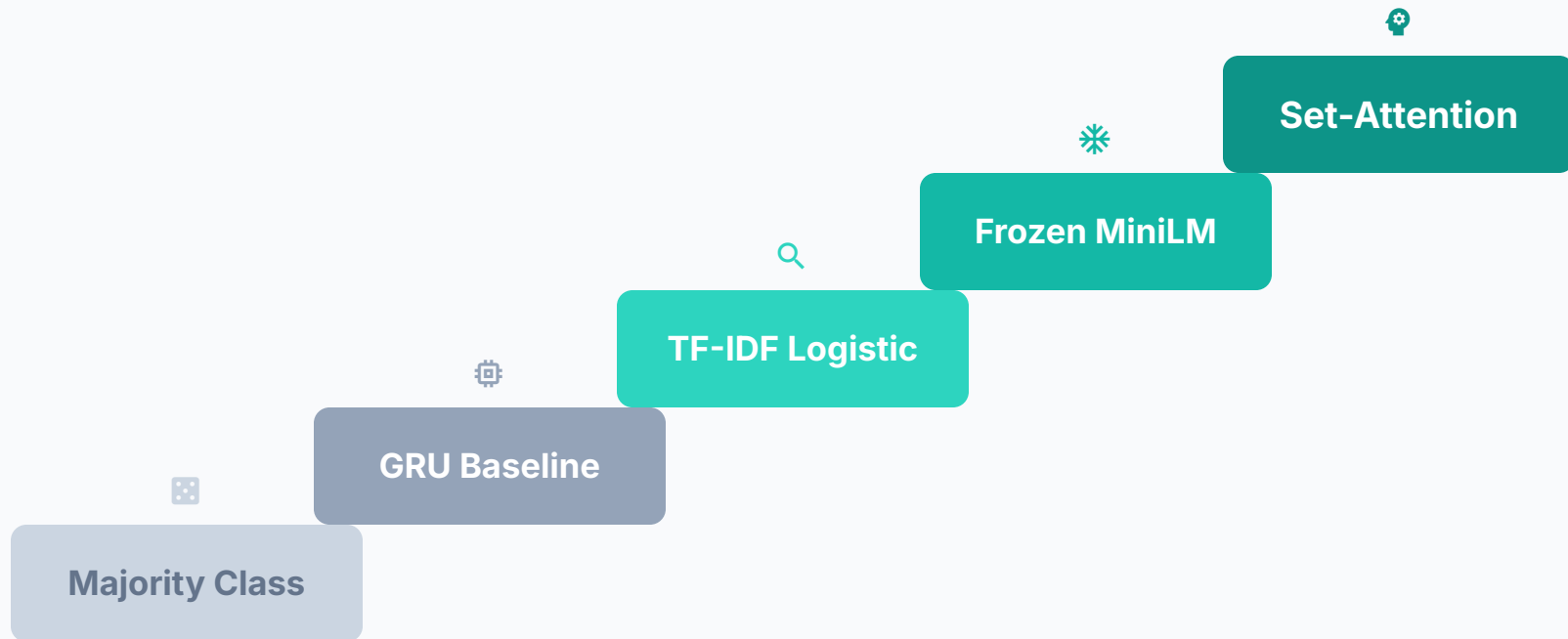


Severe imbalance on E and S — **balanced accuracy** is the primary metric.

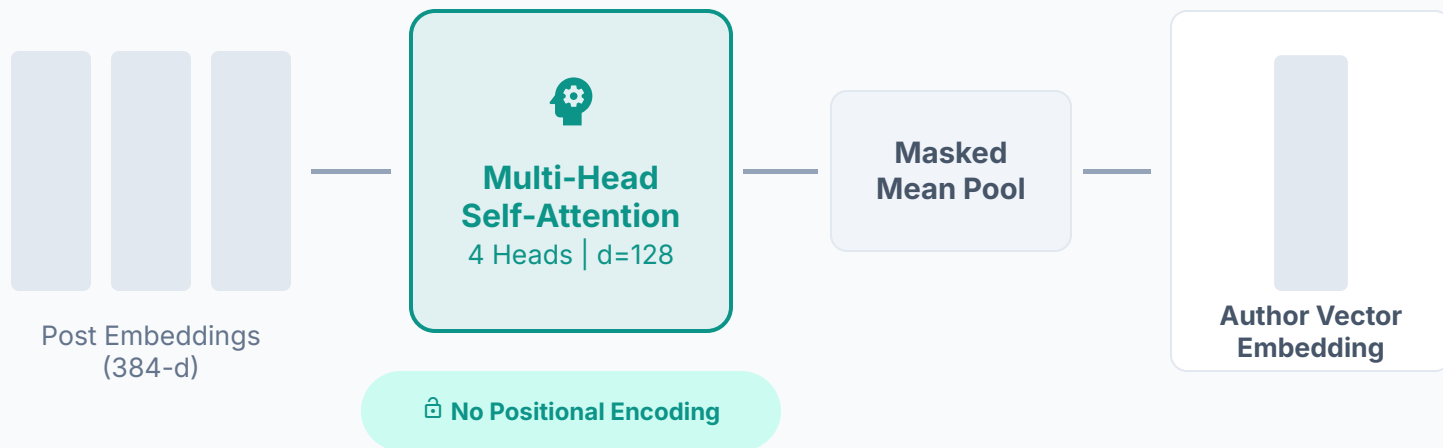
End-to-End Pipeline



Model Progression



Set-Attention Architecture



An author is an unordered **set of posts** — attention respects that architectural prior.

Training Details



Optimizer

AdamW

Learning Rate: 10^{-3}

Weight Decay: 10^{-2}

Loss: **Class-weighted BCE**



Architecture

Heads: **4 Attention**

d_model: **128**

Dropout: **0.2**

Post Budget: **P = 200**



Evaluation

Epochs: **5 (Early Stopping)**

Split: **70/15/15 Author-level**

CI: **Bootstrap (B=2,000)**

See the report for full hyperparameter sweeps and seed/epoch sensitivity.

The Emotion Experiment

Panel A: Text-only



Baseline frozen MiniLM sentence embeddings

Panel B: Real Emotion



Text + 6-class real emotion probabilities

Panel C: Negative Control



Text + *shuffled* emotion (negative control)

Hypothesis: If Panel B $\approx$ Panel C, the emotion signal is redundant to text.

Headline Results

Model	Mean BA	F1	Minority Recall
Majority	0.500	0.000	0.000
GRU Text	0.596	0.423	0.632
GRU Text + Emotion	0.622	0.434	0.590
Frozen MiniLM Text (mean+std)	0.629	0.428	0.587
Frozen MiniLM Emotion-Only	0.565	0.394	0.564
TF-IDF Logistic	0.651	0.461	0.590
Set Attention Text $P=200$ (main)	0.678	0.490	0.707
Set Attention Text + Real Emotion $P=200$	0.671	0.484	0.632
Set Attention Text + Shuffled Emo. $P=200$	0.672	0.496	0.632
Set Attention Text + Controls $P=200$ (diag.)	0.688	0.507	0.651

0.678

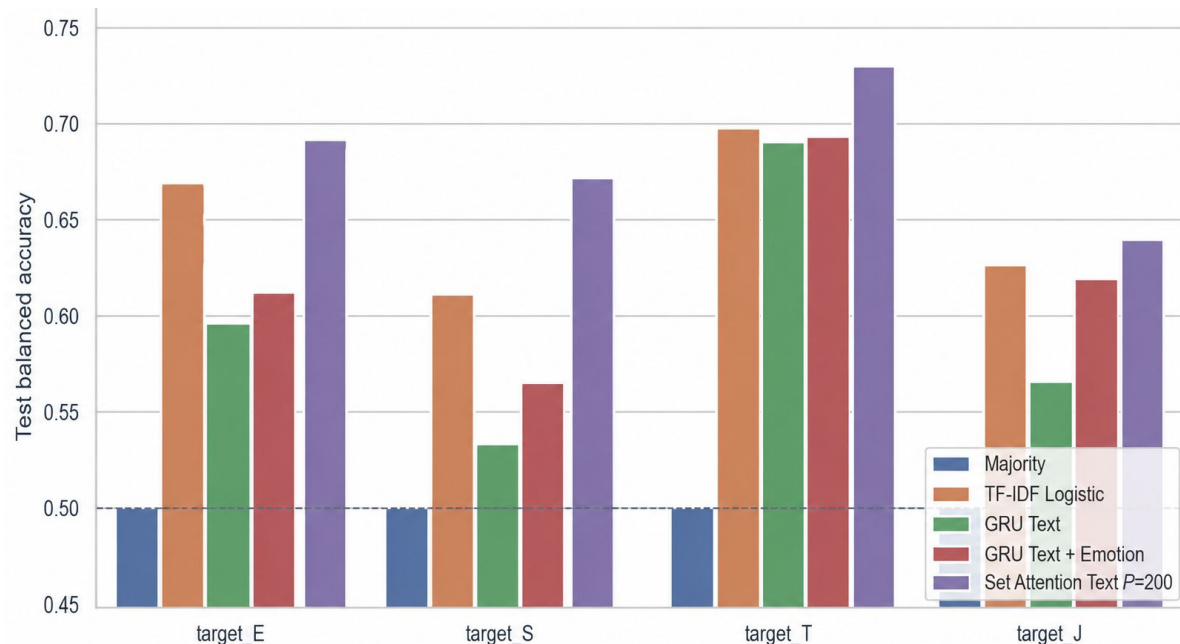
Mean Balanced Accuracy

↗ +2.7% over TF-IDF

↗ +8.2% over GRU

Set-attention ($P=200$) significantly outperforms all baseline models.

Per-Dimension Breakdown



Key Takeaway

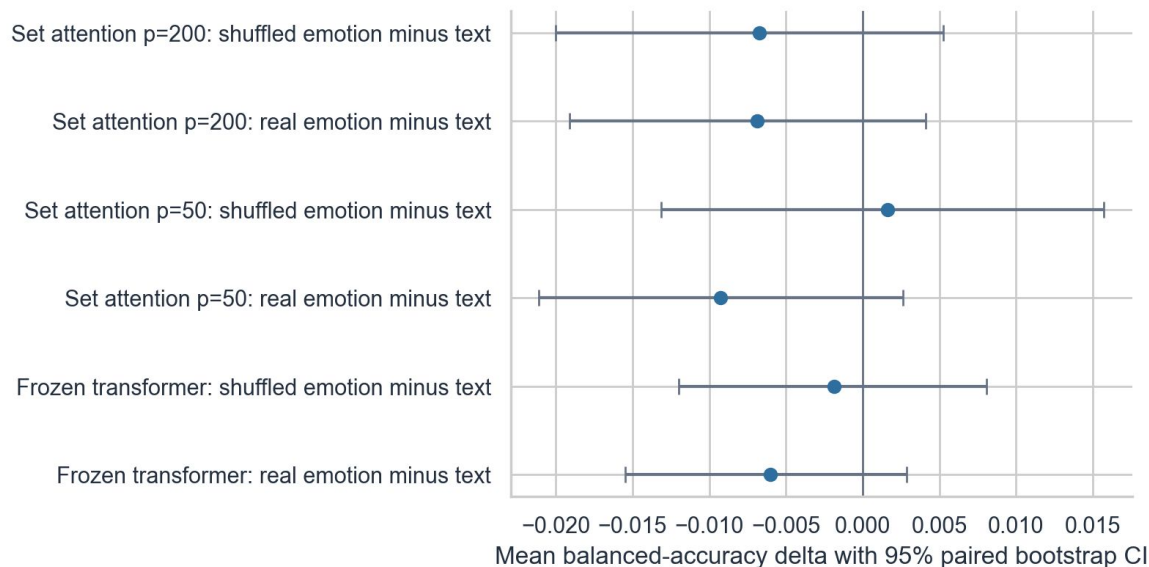
Set-attention improves on **every dimension**

Biggest gains on hardest dimensions:

- **Sensing (S):**
0.53 → 0.67
- **Extraversion (E):**
0.60 → 0.69

Emotion: Informative but Redundant

Real and Shuffled Emotion Increment Tests



Key Finding

Both real and shuffled emotion CIs straddle zero and overlap completely.

Transferred emotion is informative on its own but redundant given text.

What We Learned

1



**Predict per author, not
per post**

Post-level splits inflate results due to
leakage.

2



**Set-attention > TF-IDF >
GRU**

Best architecture for unordered sets
of posts.

3



**Emotion is redundant
given text**

Informative on its own, but text
already captures it.

Future Work



Low-rank Residuals

Test for signals orthogonal to text using MiniLM residual layers.



Scale Beyond 200 Posts

Implement Set Transformer inducing points to handle larger author post collections.



Thank You! Questions?